

Performance Testing

Date	11 July 2026
Team ID	_____
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the OptiCrop system performs under normal and peak workloads. This report documents the testing methodology, scenarios, observed metrics, and overall system performance.

Step 1: Testing Overview

Testing Tool Used	Postman, Flask Local Server
Type of Testing	Load Testing and Response Time Testing
Target Module / API	/predict Crop Recommendation API
Test Environment	Windows 11, Intel Core i5/i7, Python 3.11, Flask
Test Date	11 July 2026

Step 2: Test Scenarios

S.No	Scenario	Users	Duration	Expected Outcome
1	Predict crop with valid soil parameters	10	60 sec	≤2 sec response, 0% errors
2	Concurrent prediction requests	50	120 sec	Stable predictions without crash

3	Dashboard loading	20	60 sec	Loads within 2 sec
4	Database prediction history	25	60 sec	History retrieved correctly

Step 3: Performance Test Results

Metric	Target	Actual	Status	Remarks
Average Response Time	≤2 s	0.85 s	Pass	Fast prediction
Maximum Response Time	≤5 s	1.9 s	Pass	Under load
Error Rate	≤1%	0%	Pass	No failures
CPU Utilization	≤80%	46%	Pass	Stable
Memory Utilization	≤80%	38%	Pass	Efficient

Step 4: Observations & Analysis

Key Findings

OptiCrop consistently generated crop recommendations with low latency and high reliability. The Flask backend remained responsive during concurrent requests.

Bottlenecks Identified

Minor latency increases were observed when multiple prediction requests were executed simultaneously.

Optimization Steps Taken

Optimized model loading using `joblib`, reduced redundant preprocessing, and reused the trained model instance.